

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z2,Z3

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple **Z2 Z3:** $\frac{Z_2Z_3g_m+Z_3}{Z_2g_m+1}$

$$H(z)=\frac{Z_2Z_3g_m+Z_3}{Z_2g_m+1}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s)=(\infty, \ R_2, \ R_3, \ \infty, \ \infty, \ \infty)$

$$H(s)=R_3$$

9.2 X-INVALID-ORDER-2 $Z(s)=\left(\infty, \ R_2, \ \frac{1}{C_3s}, \ \infty, \ \infty, \ \infty\right)$

$$H(s)=\frac{1}{C_3s}$$

9.3 X-INVALID-ORDER-3 $Z(s)=\left(\infty, \ R_2, \ \frac{R_3}{C_3R_3s+1}, \ \infty, \ \infty, \ \infty\right)$

$$H(s)=\frac{R_3}{C_3R_3s+1}$$

9.4 X-INVALID-ORDER-4 $Z(s)=\left(\infty, \ R_2, \ R_3+\frac{1}{C_3s}, \ \infty, \ \infty, \ \infty\right)$

$$H(s)=\frac{R_2g_m+s\left(C_3R_2R_3g_m+C_3R_3\right)+1}{s\left(C_3R_2g_m+C_3\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s)=\left(\infty, \ \frac{1}{C_2s}, \ R_3, \ \infty, \ \infty, \ \infty\right)$

$$H(s)=R_3$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 R_3 s^2 + g_m + s (C_2 + C_3 R_3 g_m)}{C_2 C_3 s^2 + C_3 g_m s}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, R_3, \infty, \infty, \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, R_3 + \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_3 R_2 R_3 s^2 + R_2 g_m + s (C_2 R_2 + C_3 R_2 R_3 g_m + C_3 R_3) + 1}{C_2 C_3 R_2 s^2 + s (C_3 R_2 g_m + C_3)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, R_3, \infty, \infty, \infty \right)$$

$$H(s) = R_3$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \frac{1}{C_3 s}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{1}{C_3 s}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_3}{C_3 R_3 s + 1}$$

9.16

X-INVALID-ORDER-16

$$Z(s) = \left(\infty, \ R_2 + \frac{1}{C_2 s}, \ R_3 + \frac{1}{C_3 s}, \ \infty, \ \infty, \ \infty \right)$$

$$H(s) = \frac{g_m + s^2 \left(C_2 C_3 R_2 R_3 g_m + C_2 C_3 R_3 \right) + s \left(C_2 R_2 g_m + C_2 + C_3 R_3 g_m \right)}{C_3 g_m s + s^2 \left(C_2 C_3 R_2 g_m + C_2 C_3 \right)}$$

10 X-INVALID-WZ

11 X-PolynomialError